

Chapter 57 — Etiology, Diagnosis, and Management of the Undescended Testis

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- **Cryptorchidism (undescended testis, UDT)** is a common congenital anomaly that remains incompletely understood despite extensive study.
- **Descent is normally completed in the third trimester or shortly after birth**; the anatomy is well described, but the **mechanical function of the gubernaculum** — the fetal structure orchestrating descent — is still unresolved.
- **Susceptibility is complex and multifactorial**, plausibly combining genetic factors with exposure to **endocrine-disrupting chemicals (EDCs)**, potentially in combination.
- **Diagnosis is frequently made well beyond the neonatal period**, and any clinically significant long-term difference between congenital and acquired cryptorchidism remains obscure.
- **Retractile testes may represent a milder form of the disease**, conferring higher risk of acquired cryptorchidism diagnosed later in childhood.
- **Ultrasonography and other imaging are rarely if ever indicated** — they contribute little to diagnosis.
- **Hormonal therapy is no longer advocated as first-line treatment**. Surgery is the preferred treatment, with approach determined by **palpability, position, size, patient age and overall health status**.
- **Primary scrotal and laparoscopic approaches are increasingly used**.
- **Suboptimal semen quality is common in bilateral cases** and can occur in unilateral disease, though paternity is more likely in unilateral cases.
- **Testicular germ cell tumour (TGCT) risk is raised in the affected testis** and to a lesser degree in the contralateral testis; **early surgery may reduce it**.
- **Long-term outcome depends on laterality, age at treatment, and potentially the underlying aetiology**.
- **The American Urological Association (AUA) cryptorchidism guideline (2014)** provides the evidence-based framework for evaluation and management.

DEFINITIONS

TERMINOLOGY OF TESTICULAR POSITION AND ABSENCE

Term	Definition	Distinguishing feature
Normal scrotal position	Midpoint of the testis at or below the midscrotum	The reference standard against which all else is judged
High scrotal testis	Stable testis above the scrotal midpoint	Not routinely called undescended, but included as UDT in some epidemiological studies — a heterogeneous group of stable descended, retractile and unstable "gliding" testes
Undescended testis / cryptorchidism	Testis not in normal scrotal position, unilateral or bilateral	Divides clinically into palpable vs nonpalpable ; nonpalpable divides into undescended vs absent
Ectopic testis	Testis outside the ipsilateral inguinal canal, off the normal line of descent	Position rather than arrest

Term	Definition	Distinguishing feature
Vanishing / vanished testis	Testis developed then lost to vascular accident or torsion	Monorchia unilaterally; anorchia bilaterally and very rarely
Agenesis	Testis never present	Associated with ipsilateral müllerian duct persistence — the discriminator from a vanished testis
Congenital cryptorchidism	Testis outside the scrotum at birth	Baseline category
Acquired cryptorchidism	Cryptorchid testis documented as scrotal on a previous examination	Requires prior documentation
Recurrent cryptorchidism	Undescended at birth, descended spontaneously, then re-ascended	Three-stage history
Secondary cryptorchidism	Suprascrotal testis after inguinal hernia repair	May be primary maldescent or postoperative scarring
Testicular retraction	Suprascrotal testis as a complication of orchidopexy	Iatrogenic
Retractile testis	Scrotal testis retracting easily but manually replaceable and stable at least temporarily	Testes that rarely remain stable, or rest in the high scrotum, may progress to acquired cryptorchidism

Concept — *the whole taxonomy turns on two axes: where the testis is now, and what is known about where it has been. Position separates undescended from ectopic and absent; documented prior position separates congenital from acquired, recurrent and secondary.*

AETIOLOGY

Embryology of Testicular Descent and Maldescent

- Cryptorchidism is a failure of descent dependent on growth and hormonal function of the developing testis and its regulation of gubernacular growth and function.
- Descent is regulated by two Leydig cell hormones: insulin-like 3 (INSL3) acting via relaxin/insulin-like family peptide receptor 2 (RXFP2), and androgens — dihydrotestosterone (DHT) and testosterone — via the androgen receptor (AR).
- Descent also requires intact innervation of the gubernaculum by the genitofemoral nerve (GFN); downstream interactions between hormonal and neural pathways are poorly defined.
- Genetic or environmental effects altering key molecules of testicular or gubernacular development are the presumed contributors to cryptorchidism.

Differentiation of the Testis

- Gonadal differentiation is an active process in both sexes — specific multigene programmes are required for male and female determination, not a default pathway with a male override.
- SRY (sex-determining region on the Y chromosome) expression in the indifferent gonad triggers testicular development, "tipping the balance" of antagonistic male–female signalling toward the male programme.
- SRY is activated in part by steroidogenic factor 1 (SF1/NR5A1); SRY and SF1 cooperate to sustain SOX9 (SRY box-containing 9) expression in pre-Sertoli supporting cells.
- SRY specifies the Sertoli cell; SOX9, FGF9 (fibroblast growth factor 9), AMH (antimüllerian hormone), PTGDS (prostaglandin synthase) and others drive Sertoli differentiation and proliferation.

- **Establishment of the Sertoli lineage is the prerequisite for testicular cord formation.**
- **Testis patterning requires a temporal sequence:** germ cell migration into the undifferentiated gonad → migration of interstitial cell precursors from the **coelomic epithelium and mesonephros**.

Human germ cell timeline

1. **32 days post-ovulation** — gonads with somatic and germ cells first identifiable on the **medial aspect of the urogenital ridge**; gonads and genitalia still sexually indifferent →
2. **40 days** — **SRY and SOX9 transcripts appear**, followed by histological emergence of Sertoli cells →
3. **6 weeks** — **primordial germ cells (PGCs) migrate from the yolk sac** and differentiate into gonocytes →
4. **7 weeks** — gonocytes begin expressing **c-KIT**, rising through the first trimester →
5. **8 weeks** — germ cells **localised within testicular cords** →
6. **Second trimester** — three distinguishable germ cell subpopulations: **gonocytes, intermediate spermatogonia, prespermatogonia** →
7. **20 weeks** — **POU5F1/OCT4 and c-KIT lost with the gonocytes**; **MAGE-A4 immunopositivity marks emergence of prespermatogonia**.

Fetal Leydig cells

- **Arise from coelomic epithelium and perivascular cells at the gonad–mesonephric border**, forming interstitial Leydig progenitors.
- **Progenitors express AR and are themselves androgen-regulated.**
- **Development depends on multiple growth factors and on paracrine interaction with Sertoli and peritubular myoid cells**; **WT1 expression in Sertoli cells drives peritubular myoid and Leydig cell proliferation and differentiation.**
- **Fetal Leydig cells (FLCs) become active at 6–7 weeks, proliferate until about 15 weeks, and involute after 24 weeks.**
- **By the end of the first trimester the external genitalia are fully masculinised and the testis cords are established**, with both proliferating and degenerating germ cell populations present; **Sertoli and germ cell proliferation continues well into the second trimester.**

Concept — *the testis is built in a fixed order, and each step licenses the next: SRY specifies Sertoli cells, Sertoli cells make cords, and only a hormonally competent cord-bearing testis can drive its own descent. Every later failure mode is a failure somewhere along that dependency chain.*

Testicular Hormone Production

- **Fetal testicular hormone function is critical both for reproductive tract masculinisation and for descent.**
- **Hormone production is activated at discrete intervals — fetal, postnatal and pubertal — rather than continuously.**
- **Fetal Leydig cell phases (sequence): proliferation and differentiation, 7–14 weeks → maturation, to 18 weeks → involution, continuing to term.**

FETAL TESTICULAR HORMONES — SOURCE, ONSET AND PEAK

Hormone	Cell of origin	Onset	Peak / key window
Testosterone	Fetal Leydig cell	6–7 weeks, initially gonadotrophin-independent	Human chorionic gonadotrophin (hCG)-driven peak at 14–16 weeks, then responsive to fetal luteinising hormone (LH)

Hormone	Cell of origin	Onset	Peak / key window
INSL3	Fetal Leydig cell	Measurable in amniotic fluid from 11 weeks	Peaks 15–17 weeks
AMH (müllerian-inhibiting substance, MIS)	Fetal Sertoli cell	Soon after Sertoli differentiation	Müllerian duct responsive before 8 weeks; regression 9–10 weeks

Sex steroid receptors in the fetal testis

- **AR and oestrogen receptor beta (ER β) are expressed in the undifferentiated gonad at 7 weeks, role unclear.**
- **AR is present in peritubular myoid cells and some Leydig/interstitial cells but is absent from Sertoli cells in early development.**
- **This Sertoli AR absence is physiologically essential:** AR expression there would permit inappropriate androgen-induced inhibition of AMH production and precocious stimulation of spermatogenesis.
- **ER β is expressed in germ, peritubular myoid, Sertoli and some Leydig cells; ER α is limited or absent; concomitant aromatase expression suggests a role for locally produced oestrogen.**

Mini-puberty — neonatal reactivation of the hypothalamic–pituitary–gonadal (HPG) axis

- **Cross-sectional data place peak hormone levels at 1–3 months of age; LH and follicle-stimulating hormone (FSH) rise first, followed by testosterone, AMH and inhibin B (INHB); INSL3 is high in cord blood and at 3 months.**
- **Individual longitudinal urinary data place the peak earlier — at or before 1 month — and more pronounced in preterm males.**
- **Androgen insensitivity syndrome provides the mechanism:** the surge is a response to maternal androgen withdrawal, producing secondary upregulation of hypothalamic and pituitary secretion.
- **FLCs regress after birth; a neonatal Leydig population emerges at 2–3 months and regresses by the end of the first year.** As postnatal hormones wane, FLCs degenerate or regress to immature, partially differentiated cells less responsive to LH.

Postnatal germ cell events

- **Testicular volume increases postnatally, driven principally by Sertoli and germ cell proliferation;** growth correlates with HPG activation in preterm males, implying a **direct hormonal response**.
- **Sertoli proliferation continues through the first year and is a major determinant of ultimate testicular size.**
- **Prespermatogonia/gonocytes migrate to the basement membrane after birth and become undifferentiated type A spermatogonia;** adult dark (Ad) and adult pale (Ap) spermatogonia contain the spermatogonial stem cell subpopulation.
- **Migration coincides with HPG activation, so a relationship is presumed but unproven;** known regulators include **platelet-derived growth factor, thyroid hormone and cell adhesion molecules.**
- **Androgens are not established as regulators of neonatal gonocyte migration —** fetal gonocytes express AR and androgens inhibit their proliferation, but no evidence supports an androgen effect on neonatal migration or differentiation.

- **Failed migration leaves prespermatogonia/gonocytes at increased risk of apoptosis or future malignant transformation.**

Concept — mini-puberty is the therapeutic clock of this disease. *It is the one postnatal window in which the HPG axis is active, gonocytes are migrating and testicular size is being set; damage or delay here converts a positional abnormality into an irreversible germ cell deficit.*

Gubernacular Development and Testicular Descent

- **The physiology of normal descent and the aetiology of abnormal descent in humans remain incompletely understood;** knowledge is inferred from human disorders and animal models.
- **The indifferent gonad develops adjacent to, and suspended from, the mesonephros, which has replaced the pronephros by 5 weeks.**
- **A rudimentary cranial mesonephric ligament connects to the diaphragm and disappears by 13 weeks** as the mesonephros regresses.
- **There is no "cranial gonadal ligament" in humans, unlike other species.**
- **The gubernaculum attaches proximally to the epididymis, not directly to the testis.**
- **"Transabdominal descent" in the human is relative, not true migration:** fixed-section studies describe movement from **C7–T8 to T9–L3 between 5 and 7 weeks** and to sacral level by **10 weeks**; dissection studies place the testis at **L4 by 7 weeks**; fetal studies confirm the testis stays close to the internal inguinal ring from 10 weeks onward.
- **If true transabdominal descent occurs, it is unrelated to gubernacular swelling or hormonal stimulation, because it precedes them.**
- **Caudal movement of the ovary is prevented by the ovarian ligament and müllerian duct development.**

Five phases of human testicular descent (Barteczko and Jacob)

1. **Phase 1 (5 weeks) — caudal mesonephros contacts the future gubernaculum at the internal inguinal ring →**
2. **Phase 2 (7 weeks, extending 8–10 weeks) — GFN accompanies the newly formed gubernaculum (abdominal, interstitial and subcutaneous portions) and processus vaginalis; then gubernacular growth, deepening of the processus vaginalis and extension of cremaster fibres into the interstitial gubernaculum →**
3. **Phase 3 (10–14 weeks) — testicular growth with regression of müllerian ducts and mesonephros; male-specific gubernacular swelling begins at 12 weeks — after the testis is already at the internal ring →**
4. **Phase 4 (14–20 weeks) — gubernacular swelling, cremaster development and processus vaginalis migration widen the inguinal canal →**
5. **Phase 5 (20–28 weeks) — release of the distal subcutaneous gubernacular attachment and transinguinal passage; further caudal movement to birth accompanies gubernacular regression.**

What the anatomy establishes

- **Apparent transabdominal movement precedes sexually dimorphic gubernacular change and is therefore not a male-hormone-specific event.**
- **Male-specific gubernacular swelling parallels the Leydig secretory peaks — testosterone at 14–16 weeks and INSL3 at 15–17 weeks.**
- **The GFN precedes and accompanies the gubernaculum and processus vaginalis in both sexes from 7 weeks.**

- **Both smooth muscle (abdominal portion) and skeletal muscle (interstitial and subcutaneous portions) are present** in the gubernaculum, though not all observers agree; **in situ studies from 20 weeks clearly show striated muscle bundles and GFN infiltration.**
- **Whether cremaster fibres originate from abdominal wall musculature or within the gubernaculum is contested; unique innervation and transgenic mouse data favour intrinsic gubernacular origin.**

Mechanics of passage

- **Gubernacular swelling — from cellular proliferation and extracellular matrix production — is required to enlarge the inguinal canal for testicular passage.**
- **Once the canal is created, unknown mechanical factors trigger a typically rapid transinguinal passage.**
- **The gubernaculum is distally unattached during and after transinguinal passage; settling into the preformed scrotum is gradual.**
- **The scrotum develops from genital swellings visible at 7 weeks that fuse by 10–12 weeks in response to circulating DHT — the scrotum is ready long before the testis arrives.**
- **Passage into the inguinal canal rarely occurs before 22 weeks; the majority of testes are scrotal after 27 weeks.**
- **Rapid transit, early gubernacular innervation, GFN significance in rodents and the complex cremaster distribution argue that neuromuscular forces are required for descent.**
- **Microcomputed tomography in fetal rodents has further defined the role of the gubernacular bulb in formation of the processus vaginalis.**

Fig. 57.3. *The distally unattached human gubernaculum just before transinguinal passage — human fetus at 25 weeks' gestation (215-mm crown–rump length). G, gubernaculum; P, penis; S, scrotum; T, testis. (Heyns CF, J Anat, 1987.)*

Concept — the gubernaculum does not pull the testis down; it makes room and then gets out of the way. Outgrowth creates the canal and processus vaginalis while the testis stays still; regression permits the testis to follow — which is why both an outgrowth defect and a regression defect produce the same clinical phenotype.

Regulation of Testicular Descent

- **Direct evidence for hormonal control comes from animal models, primarily rodents, whose relevance is questioned on anatomical grounds.**
- **Rodent vs human anatomy: rodents have a less prominent interstitial gubernaculum, a cranial gonadal ligament, a well-developed intrinsic cremaster at the gubernacular periphery, an absent inguinal canal, and a processus vaginalis that fails to close; the processus vaginalis and cremaster are nonetheless similar enough to justify translational study.**
- **Transgenic or spontaneous rodent strains with inactivation of *Ins13*, *Rxfp2* or *Ar* are cryptorchid, as are males with abnormalities of androgen synthesis or action.**

INSL3/RXFP2 AXIS VERSUS ANDROGEN/AR AXIS

	INSL3 / RXFP2 axis	Androgen / AR axis
Classical role (Hutson, 1985)	Transabdominal phase, the "nonandrogenic factor"	Transinguinal phase, acting via the GFN
Actual relationship	Synergistic with androgen on gubernacular proliferation and growth	Synergistic with INSL3; both present before and through the swelling phase
Timing in the human	Peaks after 12 weeks , with gubernacular swelling	Peaks after 12 weeks , with gubernacular swelling

	INSL3 / RXFP2 axis	Androgen / AR axis
Additional murine role	Promotes development of the processus vaginalis	Regression of the cranial gonadal ligament — absent in humans
Cremaster	RXFP2 deletion in pre-myogenic gubernacular mesenchyme disrupts descent	AR deletion in the same cells disrupts descent

- **The two-phase paradigm is no longer cleanly separable** — apparent transabdominal descent, if it occurs, happens in the third month, before either hormone peaks.
- **Modern model: the testis remains stable as the gubernaculum swells (the "outgrowth" phase), facilitating creation of the processus vaginalis, and descends as the gubernaculum shrinks (the "regression" phase). INSL3 and androgens regulate both phases.**
- **Both hormones directly regulate myogenic differentiation of the cremaster.**

Downstream pathways

- **Myogenesis is the recurring theme** in transgenic and human syndromic models.
- **Inactivation of *Notch1* and Wnt pathway genes (*Ctnnb1*, *Sfrp1*, *Sfrp2*, *Wnt5a*, *Ror2*) markedly disrupts fetal muscle patterning.**
- **Gubernacular gene expression after DHT or INSL3 stimulation activates Wnt pathways with strong transcriptional overlap.**
- **Gubernaculum-specific *Wnt4* knockout mice develop unilateral UDT: intra-abdominal testes show raised FSH and LH with absent germ cells; inguinal testes have normal hormones but reduced fertility.**

Neural pathway

- **GFN innervation is essential to gubernacular function in rodents; the proposed mediator is calcitonin gene-related peptide (CGRP) released from GFN sensory endings.**
- **A competing theory holds that androgens act indirectly via mesenchyme along the mammary line — this conflicts with data showing functional AR within the gubernaculum itself.**
- **A CGRP role in human descent is undefined, but genome-wide association data link *RBFOX* genes — encoding RNA-binding proteins regulating CGRP production — to clinical cryptorchidism.**

Concept — hormones do not move the testis, they build and remodel the gubernaculum. *INSL3 and androgen converge on one target tissue and one process — gubernacular myogenesis — which is why two hormones with different receptors produce a single phenotype when either fails.*

KEY POINTS

- **Separate genetic pathways in gonadal determination lead to testis or ovary; SRY regulates downstream testis-determining genes in males.**
- **Differentiation of gonocytes, Sertoli cells and Leydig cells occurs between 5 and 9 weeks' gestation; the gubernaculum appears at 7 weeks and guides descent.**
- **Testosterone and INSL3, produced by Leydig cells, are critical for descent; levels peak between 14 and 17 weeks.**
- **From the second trimester, gubernacular swelling provides space for testicular passage into the scrotum between 20 and 28 weeks.**

Epidemiology

- **Cryptorchidism is one of the commonest congenital anomalies, reported in 1–9% of full-term and 1–45% of preterm newborn males; the wide ranges reflect differing definitions and examination methods.**

- **80–85% of cases are isolated (nonsyndromic).**
- **Cryptorchidism is a component of almost 500 syndromes**, causally linked to **387 genes within 414 loci** (Online Mendelian Inheritance in Man, OMIM).
- **A common subset of syndromic cases is associated with hormonal defects of the HPG axis.**

Congenital Cryptorchidism

- **Prevalence studies are confounded by** subjectivity of examination, **inclusion or exclusion of high scrotal testes**, differing populations and design.
- **Most studies support prevalence of 2–4% at birth and 1–2% at 3 months**, with frequency as high as 9% in some geographies; **no reproducible secular trend has been shown.**
- **Cryptorchidism is more common in White than non-White populations.**
- **Spontaneous descent by 3 months occurs in 45–87%** of those cryptorchid at birth in population-based prospective series.
- **Position predicts spontaneous descent: extrascrotal testes descend by 1 year in about half**, versus **high scrotal testes in the large majority (≈88%).**
- **Referral populations behave differently: spontaneous descent in a minority of those presenting before 6 months and in none presenting after 6 months** — reflecting exclusion of early neonatal descent and greater severity on referral.
- **Re-ascent (recurrent cryptorchidism) occurred in 10–22% in two large longitudinal series, usually between 1 and 5 years of age.**

Acquired Cryptorchidism

- **Once held impossible** — cryptorchidism was regarded as a congenital anomaly identifiable at birth — acquired cryptorchidism is now an established entity, first reported in 1975.
- **In practice: testes diagnosed as cryptorchid after apparent full descent at birth or in the neonatal period.**
- **It most likely represents a milder presentation of congenital cryptorchidism that escaped detection in infancy.**

CONGENITAL VERSUS ACQUIRED CRYPTORCHIDISM

	Congenital	Acquired
Age at diagnosis	At birth / neonatal period	Average 8–11 years
Position	Variable, often higher	Lower, distal to the inguinal canal; frequently lateral (ectopic) in the superficial inguinal pouch
Processus vaginalis	Often patent	Usually closed
Epididymis	Anomalies common	Usually normal (though non-fusion is high in some series)
Proposed mechanism	Failure of descent	Fibrous processus remnant tethering the cord, mobility within an open sac, or — most likely — incomplete descent from birth unmasked by somatic growth
Associations	Prematurity, low birth weight	Proximal hypospadias, abnormal germ cell development, subfertility risk, epididymal non-fusion

Retractile testes and acquired cryptorchidism

- **Retractile testes are common in normal populations and resolve with age** — supascrotal in up to about a third of 4-year-olds and a tenth of those aged 4–12 years, and **intrascrotal in all males older than 12 years** in the classic hospital series.
- **Population-based studies report retractile testes in roughly 11–15% up to 11 years and about 4% at 7–12 years.**
- **A subset become undescended over time.**
- **Selection bias is the central methodological problem: retrospective specialist series report cryptorchidism developing in up to 7–32%, whereas a large prospective birth cohort found only about 2.6% developed ipsilateral cryptorchidism, with a further 13.5% developing contralateral or secondary cryptorchidism.**
- **A prospective study of unilateral retractile testis found reduced testicular position and/or size after 5 years in the majority, measured by pubic tubercle-to-mid-testis distance and orchidometer.**
- **Pathology shared with cryptorchidism supports a continuum: reduced testicular volume, ipsilateral patent processus vaginalis, epididymal anomalies and altered cremaster innervation in retractile testes undergoing orchidopexy.**

Prevalence of acquired disease

- **True prevalence is incompletely defined**, since longitudinal documentation of prior scrotal position covers only the first few years of life.
- **In serial-examination cohorts, supascrotal testes diagnosed at birth equalled those diagnosed subsequently, and congenital prevalence fell sharply by 18 months while a comparable number of acquired cases accrued** — persistent congenital and acquired forms are of similar prevalence.
- **Retrospective series cannot separate true acquired cases from delayed referral of congenital cases.**
- **Japanese screening (neonate, 1, 3, 10 and 18 months, and 3 years): only about 2% of ascending testes had prior retractile testes, and surgery beyond 4 years of age was far lower than the 30–50% reported in Western countries** — comprehensive screening lowers acquired cryptorchidism and late surgery.
- **Serial physical examination is recommended** to determine testicular position and identify acquired cases, particularly with retractile testes.

Concept — acquired cryptorchidism is almost certainly not a testis that fell out of a normal scrotum, but a never-quite-descended testis revealed by growth. *That reframes the question from "why did it ascend?" to "why was it missed?", and justifies annual examination through childhood.*

Pathogenesis

- **The specific cause is unknown in most cases; indirect evidence indicates a multifactorial disease of combined genetic and environmental risk.**
- **The most consistent risk factors are prematurity and/or low birth weight for gestational age, familial factors relating to the maternal environment or rare genetic variants, maternal diabetes, and maternal smoking.**
- **Maternal obesity was associated in large population studies but not in meta-analysis of smaller case-control studies** — variability between populations.
- **Nonsyndromic disease shows no strong large-effect loci, but a multilocus, complex pattern of genetic susceptibility.**
- **Few consistent human environmental exposures have been identified, though animal models implicate EDCs, potentially in combination.**

Genetic Susceptibility

- **Family studies indicate heritability with polygenic, multifactorial susceptibility.**
- **Familial clustering occurs with a variable phenotype; autosomal dominance with reduced penetrance was the mode most often cited.**
- **Large registry data show moderate familial aggregation: recurrence risk ratio highest in twins, intermediate in brothers, lowest in offspring, and significantly higher in maternal than paternal half-brothers.**
- **An independent Danish study reproduced the maternal-over-paternal pattern and highest concordance in twins; restricting to surgically treated persistent cryptorchidism raised concordance in all categories.**
- **The maternal excess implies environmental and/or maternal factors, or X-linked risk alleles.**
- **Earlier studies reported several-fold increased prevalence in fathers and brothers of affected subjects.**

Candidate genes

- **INSL3 and RXFP2 are the most promising candidates, but coding variants are rare in persistently cryptorchid males.**
- **Interpretation is confounded: variants associate with a range of phenotypes, and apparently deleterious mutations occur in normal family members; few INSL3 exonic variants are clearly functional.**
- **The RXFP2 T222P polymorphism showed curtailed membrane localisation and receptor activation in vitro, but similar allele frequency in cases and controls elsewhere — not established.**
- **Exome analysis of infertile men with a history of cryptorchidism found homozygous high-impact INSL3 and RXFP2 variants correlating with bilateral disease, supporting autosomal recessive inheritance — superseding the prior assumption of dominant inheritance.**
- **Other candidates have not replicated, plausibly because most series include fewer than 100 cases and because of heterogeneity.**
- **Summary: specific causative alleles other than rare variants in INSL3, RXFP2 or NR5A1 are largely unknown, owing to confounding environmental risk, insufficient sample size, and geographic and phenotypic variability.**

Concept — cryptorchidism behaves genetically like a threshold trait, not a Mendelian one. *Strong familial aggregation with vanishingly rare causal variants, and maternal-line excess with autosomal candidates, is the signature of many small genetic contributions interacting with the intrauterine environment.*

Environmental Risk Factors

- **The maternal environment appears to contribute, but it is unclear whether through exposures, characteristics or lifestyle.**
- **Maternal smoking: mild-to-moderate increased risk, with possible increased germ cell abnormalities in cryptorchid offspring.**
- **Maternal alcohol: inconsistent.**
- **Analgesics (paracetamol, or more than one analgesic) in midpregnancy or earlier: several series suggest increased risk, supported by altered hormone production in cultured human fetal testis; recent meta-analyses judge the evidence weak.**
- **Parental occupation: not consistently associated, though horticulture-related occupations and areas of higher pesticide use may carry risk.**

Testicular dysgenesis syndrome and diethylstilbestrol

- **Testicular dysgenesis syndrome (TDS): cryptorchidism, hypospadias, infertility and TGCT** attributed to **impaired testicular function in utero**, proposed to result from **deficient fetal androgen production** caused by EDC exposure.
- Concern arose from increased cryptorchidism after maternal diethylstilbestrol (DES) exposure.
- Large cohort: risk roughly doubled overall, higher with exposure before 11 weeks' gestation and cumulative dose ≥ 5 g — early and substantial exposure carries the clearest risk.

Endocrine-disrupting chemicals — the balance

- Animal models support a link between prenatal antiandrogenic EDCs and cryptorchidism, but at exposures far exceeding clinical levels, with disputed applicability to humans.
- Species-specific sensitivity is critical: the human testis appears insensitive to phthalate or DES inhibition of steroidogenesis, but sensitive to bisphenol A (BPA).
- Measured EDC levels in placenta, breast milk or cord blood — pesticides, polychlorinated biphenyls (PCBs), dioxins, perfluorinated compounds, flame retardants (PBDEs), organotin — were not consistently higher in affected pregnancies; phthalate metabolites did not correlate with cryptorchidism.
- Positive signals: flame retardants in maternal hair, BPA in placenta, organic pollutants in adipose tissue; the Norwegian Human Milk Study (PCB-74, PCB-114, PCB-194, β -hexachlorocyclohexane); Danish registry data on maternal airborne PCB exposure.
- Mixture effects: antiandrogens with different mechanisms can cumulatively inhibit fetal reproductive development at low exposure levels.
- Assisted reproductive technology with first-trimester adjuvant corticosteroids was associated with markedly increased cryptorchidism.

INDIRECT MARKERS OF FETAL EXPOSURE

Marker	What it is	What it indicates
Anogenital distance (AGD)	Posterior scrotal margin to anal verge; sexually dimorphic, correlated with birth weight	Prenatal androgen exposure; reduced AGD correlates with cryptorchidism, hypospadias, AR CAG repeat length, phthalates, maternal analgesics, shorter breastfeeding and ESR1 variants
Placental weight	Measure of placental function; may reflect hCG production in the descent window	Reduced in cryptorchidism; early pre-eclampsia linked; consistent with the strong intrauterine growth restriction association
Maternal hCG and alpha-fetoprotein	Placental biomarkers	Not consistently associated

- **Bottom line:** potential associations with pesticides, flame retardants, BPA and organic pollutants in selected populations; cumulative mixtures of antiandrogenic and oestrogenic chemicals may raise risk in susceptible individuals; single-agent causality is hard to demonstrate.

Concept — the EDC hypothesis is a mixtures-and-susceptibility argument, not a single-agent one. *Because human and rodent testes differ in which compounds affect them, and no single exposure replicates, the coherent version requires low-dose cumulative antiandrogenic load acting on a genetically susceptible fetus.*

Hormonal Defects

- Since testicular hormones are required for descent, defective production or action may be pathogenic.

- **Direct fetal data are scant; increased variance in amniotic fluid INSL3 between 13 and 16 weeks** has been reported in fetuses later cryptorchid or hypospadiac.
- **Defects may be manifest during mini-puberty — the practical diagnostic window.**

Neonatal studies

- **Results conflict:** reduced testosterone and/or LH in some series, not in others.
- **Geography matters:** INHB and FSH differed between Finnish and Danish cryptorchid infants while testosterone did not.
- **INSL3 was reduced in cord blood but not at 3 months in persistent cryptorchidism; cord testosterone did not differ.**
- **The informative finding is ratio-based:** raised LH/testosterone and LH/INSL3 ratios indicate "compensated Leydig cell function" — subtly reduced Leydig function driving compensatory LH.
- **In the largest series (Kollin et al., 2012), hormones did not differ among unilateral, bilateral and spontaneously resolving cases and did not correlate with spermatogenesis.**

Older children

- **LH, FSH and testosterone fall to low or undetectable levels as the HPG axis becomes quiescent.**
- **INHB and AMH remain measurable throughout childhood and are the sensitive Sertoli cell biomarkers of prepubertal testicular activity.**
- **Sertoli dysfunction is increasingly evident, particularly in bilateral disease: AMH reduced roughly three times more often in bilateral than unilateral cases; gonadotrophins normal or rarely mildly raised.**
- **Other series report more raised FSH and reduced INHB but used reference standards rather than controls; another found no differences before or after hCG stimulation.**
- **Confounders include population and maternal smoking (associated with reduced INHB in offspring).**
- **Low prepubertal gonadotrophins have been read as gonadotrophin deficiency causing cryptorchidism — but the axis is inactive in childhood and gonadotrophins may be normal even in anorchia.**

Effect of surgery on hormones

- **INHB rose postoperatively in the large majority in one series; abnormal INHB was about twice as common in uncorrected as corrected cases.**
- **In other series INHB normalised in fewer than half and did not correlate with histology; raised FSH in bilateral cases normalised in most.**
- **Interpretation is limited by large intraindividual fluctuation of FSH and INHB in the first 2 years of life.**

PROPOSED HORMONAL DEFECT MODELS

	Model 1 — primary gonadal defect	Model 2 — central defect
Nature	Gonadal defect, primary or secondary to the effect of position on Sertoli function	Isolated or transient hypogonadotrophic hypogonadism
Hormonal signature	With or without hypergonadotrophic hypogonadism	Normal FSH and LH despite low INHB; poor spermatogonial differentiation with low LH; reduced Leydig response to hCG
Status	Better supported	Controversial; mini-puberty measurements tend to show increased gonadotrophin drive

- **Some cryptorchid males have HPG abnormalities, with stronger support for primary gonadal (Sertoli and/or Leydig) defects than central hypogonadism in nonsyndromic cases.**

- **No hormonal study to date precisely predicts germ cell numbers or subfertility.**

Concept — **hormonal testing is diagnostically useful only at the extremes.** *It identifies anorchia and syndromic hypogonadism during mini-puberty, but in nonsyndromic disease it neither explains maldevelopment nor predicts fertility, which is why it has no routine role.*

Syndromic Cryptorchidism

- **Undescended testes are frequent in androgen biosynthetic defects, androgen insensitivity, Leydig cell agenesis, and gonadotrophin deficiency;** generalised failure of masculinisation constitutes a **disorder of sex development (DSD).**
- **Persistent müllerian duct syndrome (PMDS) — defective AMH signalling — is associated with cryptorchidism or transverse testicular ectopia,** through hindrance by retained müllerian ducts and/or loss of AMH's proliferative effect on the gubernaculum.
- **Klinefelter syndrome (47,XXY) occurs in under 2% of nonsyndromic cryptorchidism series,** more with hypospadias; conversely, more than half of prepubertal Klinefelter patients have cryptorchidism.
- **Other rearrangements and trisomies, including Down syndrome (trisomy 21), are associated.**

CONGENITAL ANOMALIES ASSOCIATED WITH CRYPTORCHIDISM

Anomaly	Cryptorchidism	Note
Prune-belly (triad / Eagle-Barrett) syndrome	All cases	Multisystem anomalies in about 45%
Spigelian hernia	53–80%	Absent gubernaculum and inguinal canal may complicate orchidopexy
Cerebral palsy (CP)	25–54%	Multisystem in about 19%; more commonly bilateral
Arthrogryposis	38%	Musculoskeletal
Omphalocele	16–33%	Multisystem in about 80%
Imperforate anus	19%	Caudal field
Myelomeningocele	15%	Neural tube
Posterior urethral valves (PUV)	12–17%	Urological
Congenital diaphragmatic hernia (CDH)	7–30%	Ipsilateral in all cases
Gastroschisis	5–25%	Spontaneous descent in 50–55% — observe
Umbilical hernia	6%	Abdominal wall

- **Groups cluster into musculoskeletal, central nervous system (CNS) and abdominal wall/gastrointestinal defects.**
- **CNS associations are strong: CP carries by far the highest relative risk,** with lesser associations for low IQ, hypotonia, autism and other developmental abnormalities.
- **Laterality: renal and T10–S5 spinal anomalies (caudal field defects) are ipsilateral in 90%; CDH ipsilateral in all cases; CP/CNS-associated disease is more commonly bilateral.**
- **These support common developmental origins with the urogenital ridge, abdominal wall, lumbosacral spine and CNS.**

- **Extragenital anomalies may alter treatment** — Spigelian hernia technically, gastroschisis because spontaneous descent is likely.

Concept — the company cryptorchidism keeps tells you where the lesion sits. *Abdominal wall and caudal-field anomalies point to a field defect and are ipsilateral; CNS and syndromic associations point to a central or global defect and are bilateral* — and that distinction predicts whether the contralateral side is also at risk.

KEY POINTS

- Cryptorchidism occurs in 1–4% of full-term males. Spontaneous descent may occur in the first few months, usually by 6 months. Testes may also ascend over time.
- Causes are largely unknown; birth weight, gestational age, genetic and environmental factors likely contribute.
- Acquired cryptorchidism is likely equivalent to cases diagnosed at birth. It may follow confirmed scrotal position at birth or spontaneous descent, and **may be commoner with retractile testes**. The risk warrants annual testicular examination throughout childhood.

DIAGNOSIS

Examination

- Physical examination is the gold standard for diagnosing cryptorchidism.
- Positions: supine, and where necessary upright — sitting cross-legged and/or squatting/standing.
- With the patient supine, examine from the ipsilateral side of the testis in question.
- Abduction of the thighs inhibits the cremasteric reflex, which is elicited by stroking the ipsilateral inner thigh.
- Document palpability, position, mobility, size and lie, plus hernia, hydrocele, intratesticular lesions, penile size and urethral meatus position.
- Adjuncts to limit cremasteric activity: distraction, warm room and hands, liquid soap or lubricant, repeated examinations.
- Sustained gentle traction on the spermatic cord fatigues the cremaster and allows a retractile testis to remain stably scrotal — the manoeuvre that distinguishes retractile from undescended.
- Scrotal asymmetry is a useful sign in unilateral cryptorchidism.
- Contralateral testicular hypertrophy with a nonpalpable testis suggests a vanishing testis but is not diagnostic.
- 75–80% of undescended testes are palpable; 60–70% are unilateral; right-sided involvement is commoner overall but less frequent among nonpalpable testes.

POSITION OF UNDESCENDED TESTES BY SERIES TYPE

Position	Surgical meta-analysis	Single-institution series	European multi-institutional
Abdominal	34%	3–10%	8%
Near internal ring ("peeping")	12%	—	—
Canalicular	27%	16–27%	63%
Beyond external ring / prescrotal	27%	Majority distal to external ring	24%

Position	Surgical meta-analysis	Single-institution series	European multi-institutional
Superficial inguinal pouch (SIP) or ectopic	—	Moul and Belman: all subinguinal testes with lateral gubernacular attachment classed ectopic	11%

- **Position varies markedly with population and classification technique** — the reason the series disagree.

Associated genital findings and the neonatal work-up

1. If neither testis is palpable — particularly with abnormal penile development — karyotype and hormonal analysis urgently to exclude congenital adrenal hyperplasia (CAH) and pre-empt undiagnosed salt wasting →
2. Pelvic ultrasonography may identify müllerian structures →
3. Delay any planned circumcision until evaluation confirms a genetically typical male.
 - Hypospadias coexists with cryptorchidism in 12–24%.
 - Proximal hypospadias with cryptorchidism warrants chromosomal analysis — abnormalities in 32–47%.
 - Micropenis occurs in nearly half of males with anorchia from bilateral vanishing testes (testicular regression syndrome); small penis with cryptorchidism also occurs in hypogonadotrophic hypogonadism.
 - In these cases measure testosterone, LH and FSH in the first months of life — the mini-puberty window.

Concept — the examination is deciding which of three questions the child poses. *Palpable or not* determines the operation; *retractile or undescended* determines whether to operate at all; *bilateral nonpalpable with an abnormal phallus* converts the problem from urological to endocrine and urgent.

Palpable Testes

- Undescended testes lie along the line of normal descent or in an ectopic position.
- The commonest ectopic site is the superficial inguinal pouch; rarer sites: perirenal, prepubic, femoral, peripenile, perineal, transverse scrotal and contralateral scrotal.
- **Correct classification as palpable or nonpalpable is the critical step that determines further diagnosis and treatment;** every effort must be made to find the lowest position the testis can attain.

Examination technique

1. Apply downward pressure with one hand along the ipsilateral inguinal canal, from the anterior iliac spine toward the scrotum →
2. Palpate with the opposite hand for the lowest attainable position →
3. While compressing the canal, invaginate the scrotum to palpate a testis distal to the external ring or within the SIP.

Limits of clinical classification

- Difficulty lies in documenting position and distinguishing undescended from retractile testes — which may coexist.
- The gold standard is careful examination in several positions, with confirmation of incomplete descent after induction of anaesthesia.
- **Interobserver and intraobserver variation is substantial:** two examiners agreed completely on position and mobility in only a small minority.
- **Preoperative position correlates poorly with intraoperative findings.**

- Absence of an intra-abdominal testis cannot be reliably predicted by scrotal examination; boys predicted by an experienced surgeon to have nubbins had viable testes at laparoscopy.
- Variation in observed position preoperatively and postoperatively may distort assessment of outcome.

FIGURE 57.6 — AUA ALGORITHM FOR MANAGEMENT OF THE UNDESCENDED TESTIS (AMERICAN UROLOGICAL ASSOCIATION, 2014)

Step	Decision	Action
Entry	Bilateral and nonpalpable	Consult a specialist to evaluate for DSD
	Not bilateral nonpalpable; under 6 months corrected age	Reassess at 6 months corrected for gestational age
	Still undescended at 6 months corrected age	Refer to a surgical specialist
Examination under anaesthesia	Palpable — retractile	Monitor yearly
	Palpable — undescended	Orchidopexy, scrotal versus inguinal
	Nonpalpable — nubbin palpable $\pm$ contralateral hypertrophy	Consider inguinal/scrotal exploration
	Nonpalpable — otherwise	Abdominal exploration, laparoscopic versus open
Abdominal exploration	Testis identified	Orchidopexy — single stage or Fowler-Stephens (one or two stage)
	Vessels entering the internal ring	Inguinal exploration
	Blind-ending vessels	No further exploration

Concept — the algorithm is organised around two gates, age and palpability. Six months corrected age closes the window for spontaneous descent and triggers referral; palpability under anaesthesia dictates orchidopexy or exploration; and the vascular findings at laparoscopy — testis, vessels entering the ring, or blind-ending vessels — terminate it.

Nonpalpable Testes

FINDINGS AT SURGERY FOR A NONPALPABLE TESTIS

Finding	Frequency
Abdominal or transinguinal "peeping" location	25–50%
Complete atrophy or "vanishing" testis	15–40%
Extra-abdominal but nonpalpable — body habitus, testicular size, limited cooperation	10–30%

Fig. 57.7. High intra-abdominal left testis identified on laparoscopic evaluation, associated with a closed internal ring.

- If both testes are nonpalpable and not distal to the internal ring in a genetic male, at least 95% are abdominal; bilateral vanishing testis is rare.
- If neither vas nor spermatic artery is found at laparoscopy, dissect the perivesical area and retroperitoneum to the level of the kidney, because true agenesis is extremely rare.
- Perirenal or other abdominal testes may be associated with multicystic dysplastic or absent ipsilateral kidney and/or non-union of testis and epididymis.

The vanishing testis

- Thought to represent cord torsion or a vascular accident after completion of genital masculinisation but before scrotal fixation.
- Evidence: haemosiderin in excised "nubbins"; reported contralateral postnatal torsion.
- Highly predictive signs: contralateral compensatory hypertrophy, and absence of palpable intrascrotal "appendage" tissue (processus vaginalis, wolffian structures, gubernaculum).
- Contralateral volume >2 mL: sensitivity 72%, specificity 100%; contralateral length 19–20 mm accurately predicts a solitary testis.
- Useful for counselling but must not preclude surgical work-up.
- Diagnosis requires documentation of blind-ending spermatic vessels in the abdomen, inguinal canal or scrotum.

Concept — the contralateral testis is a hint, never a verdict. Hypertrophy and absent appendage tissue support counselling, but because the diagnostic endpoint is anatomical — blind-ending vessels — no measurement of the other side can end the operation early.

Hormonal Evaluation

- Anorchia shows elevated basal gonadotrophins and no response to hCG stimulation — but gonadotrophins may be unexpectedly low in mid-childhood.
- hCG stimulation testing is no longer the procedure of choice for documenting anorchia — poorly standardised, side effects, inaccuracy.
- Most cases proceed to laparoscopic or surgical exploration; hormone testing may suffice for anorchia, particularly in the mini-puberty window.
- If hormones unequivocally confirm testicular tissue, diagnostic laparoscopy may not be absolutely necessary — but proving absence remains difficult, and normal childhood gonadotrophins do not exclude anorchia; then laparoscopy is required.

ANORCHIA VERSUS CRYPTORCHIDISM — BILATERAL NONPALPABLE TESTES

Marker	Anorchia	Cryptorchidism
FSH, LH	Significantly higher	Lower
AMH, INHB	Significantly lower	Higher
Testosterone	Lower in the first year of life only	Normal
Response to hCG	Absent	Present

Concept — the hormonal diagnosis of anorchia is a mirror-image pattern (gonadotrophins up, Sertoli products down) reliably visible only during mini-puberty. Outside that window the axis is quiescent, which is why laparoscopy is the default confirmatory test.

Imaging

- Inguinoscrotal ultrasonography and magnetic resonance imaging (MRI) are not usually helpful and are not recommended for the nonpalpable testis.

IMAGING PERFORMANCE FOR THE NONPALPABLE TESTIS

Modality	Sensitivity	Specificity	Comment
Ultrasonography, overall	45%	78%	Inadequate for the decision it would inform
Ultrasonography, selective, inguinal testes	95–97%	—	Examination under anaesthesia gives the same information

Modality	Sensitivity	Specificity	Comment
MRI (meta-analysis)	65%	100%	Better for inguinal than abdominal; sedation needed; unlikely to change management
Magnetic resonance angiography (MRA)	96% originally; 57% in younger boys	—	Failed replication
Any modality, vanishing testis	None reliable	—	—

- One possible MRI indication: an ectopic abdominal testis not localised by laparoscopy.
- Imaging remains common: about half of referred boys had ultrasonography, which delayed surgery by about 3 months and was accurate only about half the time; imaging matched examination in under 60% and surgery in under two-thirds, with median referral age 24 months.
- Counterargument: ultrasonography may find an inguinal testis or remnant and avoid laparoscopy — not standard practice and controversial.
- Imaging is not indicated: limited accuracy, unnecessary health care use, and it does not obviate definitive surgery.

Concept — imaging fails not because it cannot see testes, but because no result changes the operation. A positive scan does not avoid surgery and a negative scan does not exclude a testis; the only measurable effect is delay.

Diagnostic Laparoscopy

- Diagnostic laparoscopy, followed by laparoscopic orchidopexy if an abdominal testis is present, is the preferred approach, preceded by examination under anaesthesia.
- Observe: size and position of spermatic vessels and vas; testicular size, quality and position; patency of the internal ring.

LAPAROSCOPIC FINDINGS AND THEIR MEANING

Finding	Interpretation	Caveat
Closed internal ring + blind-ending spermatic artery and vas	Confirms an abdominal vanishing testis	The only unequivocal combination
Hernia present	Frequently but not always a viable abdominal or distal testis	Not diagnostic alone
Atretic cord through a closed ring	Suggests a distal vanishing testis	Subjective
Normal-appearing vessels	Viable or vanishing testis	Non-discriminating
Apparent abdominal blind-ending vessels	May coexist with a distal or ectopic abdominal testis	The dangerous false negative

- If laparoscopy does not unequivocally localise the testis or blind-ending artery, further exploration is needed — laparoscopically with additional ports if desired.

Excision of the remnant

- Germ cells and/or tubules are present in 5–15% of excised remnants; malignancy risk unknown; a single case of carcinoma in situ in a remnant has been reported.
- Excise when spermatic vessels traverse the internal ring, to confirm no viable or atrophic testis, since laparoscopic appearances may deceive.
- Initial scrotal exploration can be considered with a palpable scrotal nubbin and contralateral hypertrophy (testicular length ≥ 1.8 cm); laparoscopy is then needed in only about a quarter.

- **Against scrotal-first:** futile if the vanishing testis is intra-abdominal, and in testicular-epididymal dissociation the "nubbin" may be epididymis.
- **Laparoscopy** is the procedure of choice unless a prominent scrotal nubbin is palpable with other signs of monorchism.

CONTRALATERAL FIXATION OF A SOLITARY TESTIS

For fixation	Against fixation
Eliminates the risk of losing the only testis	Prenatal torsion does not imply postnatal contralateral torsion
A contralateral bell-clapper deformity (incomplete fixation to the tunica vaginalis) may be present	Bell-clapper anomaly contralateral to vanishing testes is rare; torsion risk minimal
Empirically recommended by some surgeons	AUA algorithm: close without further action if blind-ending vessels are found

Concept — laparoscopy answers a vascular question, not a positional one. *The testis may be hidden, but the vessels cannot be; every branch terminates on what the spermatic artery does, which is why an equivocal vascular view mandates further exploration rather than closure.*

Associated Pathology

Testicular Maldevelopment

- Systematic germ cell counts established arrested spermatogonial development with progressive loss in cryptorchid testes.
- Germ cell numbers decrease from after infancy and fail to increase with age, in cryptorchid and — to a lesser degree — contralateral descended testes.
- Abnormal histology is reported in 38–98% of affected testes, reflecting exclusion of hormone responders from biopsy, differing methods, and use or non-use of age-corrected norms.

NORMAL SPERMATOGONIA PER TUBULE (S/T) BY AGE

Age	Mean S/T
Birth	2.5
1 year	2.2
2–3 years	1.2 — nadir, about half the values before and after
4 years	2.0
5 years to early puberty (11 years)	2.5, steady

- The 2–3 year nadir requires age-specific reporting.
- Autopsy data from normal boys aged 0–1.9 years show gradual reduction with no secondary rise — the difference reflecting autopsy versus biopsy sampling and non-overlapping age ranges.
- Within- and between-biopsy variability exists in single testes.

What the histology establishes

- Abnormal germ cell development is often present after early infancy in cryptorchid testes, and to a lesser degree in their descended partners.
- Pathology is similar across true ectopic, SIP, abdominal and ascending testes, though some studies show greater severity with abdominal location.
- Germ cell depletion also occurs in myelomeningocele, PUV and prune-belly syndrome.

- **Secondary cryptorchidism after hernia repair shows similar findings, suggesting primary cryptorchidism, though malposition may contribute.**
- **Higher germ cell counts correlate with less interstitial fibrosis, younger age at surgery, and palpability.**
- **Testicular volume correlates with germ cell count in both testes but cannot predict it individually.**

Sertoli and peritubular myoid cells

- **Disrupted prepubertal Sertoli morphology, failure of pubertal maturation, and reduced Sertoli number after 4 months of age.**
- **Type IV collagen — from Sertoli and myoid cells, possibly for cell-to-cell communication — is reduced in basement membranes of undescended and contralateral testes.**

LANDMARK STUDY — KOLLIN ET AL., RANDOMISED ORCHIDOPEXY AT 9 MONTHS VERSUS 3 YEARS

Endpoint	Orchidopexy at 9 months	Orchidopexy at 3 years
Sertoli cell count	Significantly higher, still higher at 4 years	Lower
Germ cell number at biopsy	Preserved	Markedly reduced
Postoperative testicular growth	Superior	Inferior
INHB	Correlates with testicular volume (Sertoli number)	Correlates with testicular volume

- **The undescended testis is smaller at birth and grows less well than the scrotal testis, even after spontaneous descent — an intrinsic defect.**
- **Conclusion: cryptorchidism has primary and secondary effects; an extrascrotal position is detrimental even in infancy.**

Gonocyte-to-spermatogonium transformation

- **The gonocyte-to-spermatogonium ratio is normal at about 1.5 months, but gonocyte disappearance and Ad spermatogonial appearance are delayed in the undescended testis.**
- **Cryptorchid testes lacking Ad spermatogonia: 7–85%, geographically variable.**
- **Reduced Ad spermatogonia per tubule (Ad/T) also occurs in the contralateral testis and in acquired cryptorchidism — evidence that acquired testes were never truly descended.**
- **Primary spermatocyte appearance at 4–5 years is delayed.**
- **As Sertoli cells provide paracrine signalling for spermatogonial differentiation, defects may lie in Sertoli cells, germ cells, or both.**

Concept — the cryptorchid testis is abnormal before it is misplaced, and misplacement then makes it worse. *Smaller size at birth, contralateral involvement and identical pathology across positions argue for an intrinsic defect; better growth and Sertoli counts after early surgery show position adds a second, preventable injury — and only the second is under the surgeon's control.*

Anomalies of the Epididymis, Processus Vaginalis, and Gubernaculum

- **Abnormal epididymal attachment: 16–75%, partly because identification is subjective.**
- **In decreasing frequency: non-fusion of caput and/or cauda → elongation and/or looping → atresia.**
- **Anomalies correlate with severity of cryptorchidism and degree of processus closure.**
- **Commoner in congenital (37–60%) than acquired (11–31%) disease.**
- **Complete non-union of testis and epididymis/vas is rare, favours abdominal testes, and does not correlate with histology.**

- Effect on sperm transport and subfertility is undefined.

PROCESSUS VAGINALIS AND GUBERNACULUM IN CRYPTORCHIDISM

Finding	Frequency
Patent processus — unilateral undescended testis	87%
Patent processus — bilateral undescended testes	71%
Aberrant gubernacular attachment lateral to the scrotum	66–75%
Patent processus — ascending (acquired) testes	45–50%

- Inguinal hernia is commoner in family members of cryptorchid males.
- Tunica and processus anomalies predispose to torsion and clinical hernia respectively.
- Torsion of an undescended testis can occur at any age and may mimic incarcerated inguinal hernia; risk is higher than in scrotal testes, particularly with developmental disorders; delay is common and testicular loss risk high.
- Torsion with loss can occur in infancy during observation for spontaneous descent.

Concept — the gubernaculum, processus and epididymis fail together because they are regulated together. *Lateral gubernacular attachment, patent processus and non-fused epididymis are three readings of one developmental defect, which is why they track severity together.*

Other Testicular Anomalies Associated With Cryptorchidism

- Polyorchidism, splenogonadal fusion and transverse testicular ectopia (TTE) each have about 100–150 reported cases; abdominal cryptorchidism is common, so laparoscopy is useful for diagnosis and treatment.

POLYORCHIDISM, SPLENOGONADAL FUSION AND TRANSVERSE TESTICULAR ECTOPIA COMPARED

	Polyorchidism	Splenogonadal fusion	Transverse testicular ectopia
Definition	Supernumerary testis	Continuous or discontinuous fibrous union of spleen and gonad; much commoner in males	Both testes descend to the same side
Site	More often unilateral and left; testes scrotal 75%, inguinal 20%, abdominal 5%	Continuous form 55%; cryptorchidism ~30%, mostly abdominal and bilateral (59%); left 65%, right 26%	Inguinal hernia with contralateral nonpalpable testis; both may lie in one hemiscrotum
Proposed cause	Duplication or division of the genital ridge ± wolffian duct	Field defect with aberrant splenic cell migration at 5–8 weeks	Hindrance by fused wolffian derivatives or persistent müllerian ducts, or a primary gubernacular defect
Presentation	Often asymptomatic; found at orchidopexy or hernia repair (24–30%); mass, pain (7%), torsion (13–15%), müllerian remnants (7–9%), tumour	Incidental at orchidopexy or hernia repair, or scrotal swelling from reactive change in splenic tissue; discontinuous form presents as a testicular mass	Hernia with contralateral nonpalpable testis

	Polyorchidism	Splenogonadal fusion	Transverse testicular ectopia
Associations	Tumour risk probably from associated cryptorchidism or PMDS	Continuous form: limb defects, micrognathia, microglossia, anal atresia, pulmonary hypoplasia; less often cardiac, cleft palate, myelomeningocele	PMDS in 20–50%
Imaging	Sonographically normal supernumerary testis	Discontinuous: hyperechoic mass with multiple hypoechoic nodules and obvious vascularity	—
Management	Observe normal scrotal testes; orchidopexy if undescended with intact ductal drainage	Recognise at orchidopexy; avoid unnecessary orchidectomy	Laparoscopy; with vasal fusion, mobilise ipsilaterally and place via a trans-septal approach

KUMAR CLASSIFICATION OF POLYORCHIDISM — BY DUCTAL DRAINAGE

Group	Type	Anatomy
A — vasal drainage	A1	Separate epididymis and vas
	A2	Separate epididymis
	A3	Shared epididymis and vas
B — no vasal drainage	B1	Epididymis present
	B2	No epididymis or vas

- Management must also account for testicular position, size and attachments.
- Testicular malignancy with splenogonadal fusion is rare and probably related to cryptorchidism, not the splenic tissue.

PERSISTENT MÜLLERIAN DUCT SYNDROME — LARGEST SERIES

Feature	Finding
Bilateral cryptorchidism	59%
Unilateral cryptorchidism with contralateral hernia (hernia uteri inguinalis)	19%
Transverse testicular ectopia	22%
Mutation in AMH or its type II receptor (AMHR2)	88%
Adult infertility	≥80%
Adult testicular malignancy	33%

- In *Ins1/3*-null mice, loss of gubernacular attachment, TTE and/or torsion occur.

Concept — all three anomalies matter because they change the operation, not the diagnosis. Each presents at orchidopexy or hernia repair, and in each the error to avoid is removing tissue that should be preserved or missing a second testis on the wrong side.

KEY POINTS

- About 80% of undescended testes are palpable, and 60–70% are unilateral.
- A patent processus vaginalis (indirect inguinal hernia) and epididymal anomalies are commonly associated with cryptorchidism.

- Subsets of cryptorchid males have subtle HPG abnormalities during mini-puberty and/or reduced Sertoli cell hormones (AMH or INHB).
- Imaging is rarely useful for the nonpalpable testis. Diagnostic laparoscopy is the procedure of choice.

GENERAL GUIDELINES FOR MANAGEMENT

- Indications for surgical correction: optimise testicular function; reduce and/or facilitate diagnosis of malignancy; cosmesis; prevent hernia and torsion.
- Except in complex medical illness or the postnatal period, treat once the diagnosis is confirmed.
- The AUA algorithm governs the approach, with similar guidelines from the Canadian Urological Association and the European Association of Urology (EAU).

Timing

- Observe for the first 6 postnatal months to allow spontaneous descent.
- If descent does not occur, operate after 6 months, preferably by 12 months of corrected gestational age; 18 months is the guideline outer limit.
- Rationale:
 1. Spontaneous descent is unlikely in full-term males after 2–6 months
 2. Testicular growth is restored after orchidopexy at 9 months compared with 3 years
 3. Abdominal orchidopexy may be easier in young infants, soon after mini-puberty
- Prematurity: observe for 6 months beyond the expected date of delivery, or — if position is marginal — until 1 year of age.

The implementation gap

- Intervention is delayed in many children worldwide — Sweden, Australia, the United Kingdom, Germany, China, Canada and the United States.
- Sweden: only about two-thirds of diagnosed boys underwent surgery, median age fell from 6.3 to 3.4 years, under 5% were operated in the first year, and orchidopexy by 2 years rose from 10% to 30%.
- The majority are still operated beyond 2 years; United States compliance has improved since the AUA guideline but remains a minority.
- Reasons: provider education, local resources, socioeconomic factors, and recurrent cryptorchidism; studies cannot separate delayed diagnosis from delayed treatment.
- Annual scrotal examination is needed after spontaneous descent in infancy, because of re-ascent risk.

Observation of acquired cryptorchidism

- Dutch observational series reported spontaneous descent in 57% and 75% of acquired cryptorchid testes, most by midpuberty; in a larger cohort 90% of high scrotal and 64.5% of inguinal/nonpalpable testes descended at a mean age of 12.9 years.
- Descended testes had volumes closer to normal than operated testes.
- Confounded: high scrotal testes dominated those descending, retractile testes were included, and some required surgery for hernia, pain or torsion.
- Oligospermia did not differ significantly between operated and observed cases, but studies were non-randomised with heavy loss to follow-up.

- **Without paternity and malignancy outcome data, observation cannot be recommended for acquired cryptorchidism, particularly bilateral.**

Concept — the timing rule is a compromise between two clocks running in opposite directions.

Spontaneous descent is possible until about 6 months and argues for waiting; germ and Sertoli cell loss accrues from infancy and argues for operating — the window between 6 and 12 months corrected age is where the curves cross, and prematurity shifts the clock, not the logic.

MEDICAL MANAGEMENT

- **Historical indications: distinguish retractile from undescended testes; stimulate descent; stimulate germ cell maturation; facilitate abdominal orchidopexy.**
- **hCG "success" in retractile testes varies from 58% to 100%, depending on age, retractility, diagnostic accuracy and dose — hCG does not reliably distinguish retractile from cryptorchid testes, nor remove the need for serial examination.**

Therapeutic Luteinising Hormone–Releasing Hormone and/or Human Chorionic Gonadotrophin

HORMONAL AGENTS USED TO INDUCE DESCENT

Agent	Route / class	Efficacy for descent	Verdict
Luteinising hormone–releasing hormone (LHRH)	Intranasal; gonadotrophin-releasing hormone analogue	About 20%; statistically superior to placebo but not clearly clinically significant	Not recommended
hCG	Intramuscular; LH-receptor agonist	Limited; never tested in placebo-controlled trials because of route	Not recommended
hCG, other uses (acquired cryptorchidism, palpability, abdominal testis)	—	Limited	Not recommended

- **Hormonal therapy to stimulate descent is not recommended**, despite over 70 years of use on the premise that androgens promote descent.

Effect on germ cells

- **Against:** hCG associated with increased germ cell apoptosis and lower adult testicular volume; prior hCG or LHRH associated with reduced germ cell counts.
- **For:** randomised LHRH before surgery raised germ cell counts, more so under 2 years of age.
- **Neutral:** LHRH non-responders versus untreated boys, all operated before 2 years, showed no differences in semen or endocrine function.
- **Whether hormones help or harm germ cells is unclear, and research is unlikely as the approach is no longer recommended.**

Adjuvant Hormonal Therapy

- Hadziselimovic first advocated adjuvant or neoadjuvant LHRH agonist therapy to improve fertility potential.
- **Rationale:** "defective mini-puberty" or relative hypogonadotrophic hypogonadism impairs gonocyte-to-Ad spermatogonium transformation, restorable later by gonadotrophin stimulation.

- **Unvalidated:** no study correlates mini-puberty hormones with spermatogonial development; HPG defects are not confirmed in most cryptorchid infants; and age at scrotal placement — not LH, testosterone, FSH or INHB — predicted germ and Sertoli cell counts in the Kollin randomised study.

Regimen and supporting evidence

- **Buserelin, low-dose and long-term** — 10 or 20 micrograms intranasally on alternate days for 6 months — dosed to avoid HPG downregulation, with orchidopexy.
- **Non-randomised support:** higher germ cell counts after buserelin; improved histology on repeat biopsy after buserelin or nafarelin; **higher germ cell counts with buserelin in a randomised comparison against placebo plus hCG and surgery alone**; higher adult sperm counts after buserelin — **but in small, unmatched groups with atypically low surgery-only counts.**

RANDOMISED PILOT STUDIES OF ADJUVANT LHRH IN BILATERAL CRYPTORCHIDISM

	Thorup et al., 2018	Vincel et al., 2018
Population	Bilateral, 7 months–3.5 years, low S/T; position not stated	Bilateral, 7–55 months, absent Ad spermatogonia
Intervention	LHRH on alternate days for 16 weeks from 3 months postoperatively	Buserelin 10 micrograms intranasally on alternate days for 6 months
Treated group	S/T improved in some; Ad/T normalised in three	S/T ≥ 0.4 in four; all developed measurable Ad/T, still below normal (≥ 0.1)
Control group	Two normalised S/T and Ad/T; mean S/T higher with treatment	S/T > 0.4 in one; no Ad spermatogonia
Limitation	Tiny numbers, unstated position	Tiny numbers; more abdominal testes in controls

- **Pilot data suggest a potential role in selected bilateral cases, but which patients benefit is unclear;** as Ad/T > 0.2 predicts fertility better than S/T, translation to adult fertility is uncertain.
- **Biopsy is not routine**, so candidates cannot usually be identified; **gonadotrophin insufficiency was present in only about a quarter of those with low germ cells per tubule**, so hormones cannot substitute for histology.
- **Systematic appraisal: high or unclear risk of bias in most domains**, with wide variation in age, sample size, agent, dose, timing, outcome and blinding, and hormone use in some controls.
- **Adjuvant therapy is not widely used; large standardised trials in bilateral cryptorchidism reporting paternity are needed.**

Concept — hormonal therapy fails on both claims for the same reason: **position, not endocrinology, is the modifiable variable.** *Descent is not androgen-limited after the fetal window, and germ cell development tracks age at scrotal placement rather than any hormone — leaving only an unproven adjuvant role in bilateral disease with absent Ad spermatogonia.*

SURGICAL MANAGEMENT

- **Surgery once failure of spontaneous descent is confirmed is standard.**
- **Retrospective early-versus-late studies are mixed; the prospective randomised Kollin study supports surgery in the first year for congenital cryptorchidism on growth and histology — long-term outcome data are lacking.**
- **Median age at orchidopexy remains 1.5–4 years in many series; only 18% are operated by 1 year of age in a large national database.**

- **Causes of lateness:** delayed referral, new acquired diagnoses, prematurity or morbidity, excessive observation, undetected re-ascent, and difficulty distinguishing retractile testes.
- **Orchidectomy** is an option in pubertal and postpubertal males, especially if the testis is abdominal or hard to mobilise and the contralateral testis is normal — spermatogenesis is usually poor and carcinoma in situ and torsion risks exist; a prosthesis may be placed at the same sitting.

Palpable Testes

Inguinal Orchidopexy

1. Re-examine after induction to confirm palpability and lowest position →
 2. Low transverse incision in Langer's lines at or below the inguinal crease, superolateral to the pubic tubercle →
 3. Search the superficial inguinal pouch during subcutaneous dissection →
 4. Open the external oblique from the external ring, sparing the ilioinguinal nerve →
 5. Record position relative to the canal →
 6. Isolate the cord; dissect distally to the gubernacular remnant; transect the gubernaculum distal to the sac to protect a long-looping vas →
 7. Incise the internal spermatic fascia longitudinally to free an intact sac without skeletonising vas and vessels (or open the sac over the testis and free its posterior wall) →
 8. Mobilise the sac to the internal ring and suture-ligate →
 9. Incise internal spermatic and transversalis fascia at the ring for retroperitoneal mobilisation if needed →
 10. Further length: divide lateral fascial bands → cranial retroperitoneal dissection → Prentiss manoeuvre (medial transposition beneath the epigastric vessels) → cranial extension of the incision →
 11. Transverse scrotal incision and subdartos pouch; tunnel anterior to the pubis →
 12. Pass the testis through the dartos without twisting the cord →
 13. Excise appendages, inspect the epididymis, record three-dimensional testicular and contralateral volumes →
 14. Fix by tension-free dartos closure around the cord incorporating the tunica vaginalis edge; additional tunica-dartos sutures or tunica albuginea sutures if needed →
 15. Close with absorbable sutures; local infiltration, ilioinguinal block or caudal anaesthesia (caudal especially for bilateral or combined penile surgery).
- If the testis cannot reach the scrotum, a two-stage procedure is an alternative to orchidectomy, which is reserved for visibly abnormal or atrophic testes, postpubertal patients, or insufficient vasal length.
 - Transalbugineal fixation is unnecessary: 98.7% success in 1104 orchidopexies without it; it carries a theoretical inflammatory or vascular risk.

Testicular biopsy

- Advocated by Hadziselimovic for fertility prognosis; controversial and not routine outside research, as it does not change treatment and may injure the testis.
- Counter-position: consider in bilateral cryptorchidism to find absent Ad spermatogonia (gonadotrophin candidates), and for carcinoma in situ in males older than 13 years.
- Risk is theoretical — no increase in microlithiasis or antisperm antibodies long-term.
- Biopsy is indicated in sexual ambiguity or clinical testicular dysgenesis.

*Complications and follow-up***ATROPHY AND FAILURE AFTER ORCHIDOPEXY**

Series	Atrophy / failure
Pre-laparoscopic-era literature review	~15% atrophy or nonscrotal position overall; 24% abdominal or peeping vs 10% distal to the internal ring; worse after 6 years
Single institution, 18 years of open orchidopexy	Retraction <2%, slightly higher in older boys; atrophy undefined
Single institution, 418 orchidopexies	Atrophy 1.9%; nonscrotal 10.3%; more complications with younger age; all acquired cases successful

- The age effect differs by era and case mix: older age worsened outcome historically, whereas recent data show earlier orchidopexy is more technically demanding, making surgeon expertise relevant.
- Minimum 6 months' follow-up for position and size; long-term counselling on fertility, malignancy and self-examination.
- Post-orchidopexy torsion is very rare, minimised by extravaginal subdartos fixation.
- Complete intrascrotal atrophy needs no further surgery, but a prosthesis should be offered — at least 6 months after any scrotal procedure or after puberty, via an inguinal approach, fixed to the dartos with purse-string non-absorbable fascial closure; complications under 5%; cryptorchid males request prostheses less often than those with post-pubertal loss.

Reoperation

- Indicated for a nonscrotal testis after orchidopexy.
- Prescrotal testis: a primary scrotal approach may suffice.
- Redman: lateral approach to the cord after mobilising external oblique and cremasteric fasciae, avoiding scarred anterior layers.
- Cartwright: intracanalicular cord mobilisation with a patch of external spermatic fascia.
- Correct a patent processus and mobilise the cord retroperitoneally in high recurrent cryptorchidism.
- Results match primary surgery, though vascular and vasal injury risk is theoretically higher.

Trans-scrotal Orchidopexy

- A primary scrotal approach can be used for palpable testes; some reserve it for testes near or drawable into the scrotum.
- Sequence: re-examine → superior scrotal border incision (Bianchi and Squire), or low transverse or midline scrotal incision for testes drawable into the scrotum → deliver the testis → mobilise the sac and cremaster "high above the inguinal canal" → fixation (often transalbugineal) → subdartos pouch.
- Some require inguinal conversion for sac ligation or cord length, though the sac can often be ligated above the internal ring through the scrotal incision.

OUTCOMES OF TRANS-SCROTAL ORCHIDOPEXY

Measure	Figure
Additional inguinal incision	3.5–4.4%
Early complications	1.6%
Reoperation	4.9%
Atrophy / hypotrophy	0.3–0.6%
Hernia present	30%

Measure	Figure
Overall complications	3% — recurrence 0.6%, haematoma 1.4%, wound infection 0.8%, one vasal injury
Randomised trial vs inguinal	Less pain and analgesic use; inguinal conversion in 24%

- Efficacy and complications are similar to inguinal orchidopexy; the scrotal approach is increasingly advocated as first choice for all palpable testes.

Concept — the inguinal and scrotal approaches differ in where the surgeon starts, not what must be achieved. Both require cord mobilisation to a dependent scrotal position and end in a subdartos pouch; the scrotal route defers the inguinal incision until it proves necessary, which for most palpable testes it does not.

Abdominal Testes

- Decide: open or laparoscopic; one- or two-stage; with or without spermatic vessel transection.
- Orchidectomy for poorly viable or higher-risk testes — pubertal or postpubertal patients, very small or dysgenetic testes — best done laparoscopically.

Open Transabdominal Orchidopexy

- Access via an extended inguinal incision opening internal oblique and peritoneum, or a higher incision medial to the pubic tubercle with a preperitoneal approach.
- Jones and Bagley (sequence): muscle-splitting approach to the internal ring → open peritoneum → deliver testis → free vas and vessels from peritoneum → tunnel to scrotum → secure as for inguinal orchidopexy. Success 95%.

Laparoscopic Orchidopexy and Fowler-Stephens Orchidopexy

- Laparoscopy has been the procedure of choice for abdominal orchidopexy since the mid-1990s, with durable high success.
- Primary versus Fowler-Stephens depends on vas and vessel length, looping ducts and age; observed position correlates poorly with ultimate cord length.

Technique

1. Re-attempt palpation after induction (laparoscopy may also serve high canalicular testes) →
 2. Decompress bladder and stomach; 5-mm supra- or infraumbilical trocar for a 0- or 30-degree lens; open Hasson or Bailez entry in children →
 3. Carbon dioxide pneumoperitoneum, maximum 8–12 mmHg →
 4. Inspect both internal rings; assess testicular size and position →
 5. Additional 2–3-mm lower-quadrant trocars triangulated to the ipsilateral ring (midclavicular at umbilical level for bilateral), or 3-mm instruments through stab incisions →
 6. Mobilise structures distal to the ring — epididymis/vas and gubernaculum (gubernaculum as a handle, minimal cautery) →
 7. Incise peritoneum lateral to the vessels and distal to the vas →
 8. Mobilise vessels proximally, preserving the collateral supply between vas and spermatic vessels — so Fowler-Stephens remains possible →
 9. Bring the testis through a new hiatus medial to the inferior epigastric vessels and lateral to the medial umbilical ligament, or through the internal ring, via a trans-scrotal clamp or port →
 10. Divide peritoneal attachments over the cord under traction, avoiding excessive tension that risks vessel avulsion.
- Reaching the opposite internal ring is an unreliable test of adequate length.

- Some testes reach only the upper scrotum; long-term adequacy is unclear.
- Formal closure of the dissected ring is unnecessary — no hernias at 41–50 months despite closure in only about half.
- A contralateral patent processus appears in about 9%; repair is recommended by some, but its value in preventing hernia is questionable.

Fowler-Stephens principles

- For testes not within 2–4 cm of the internal ring.
- A long-looping vas helps but is not required.
- Now usually laparoscopic vessel clipping, with mobilisation in the same sitting or staged at 6 months.
- Leave peritoneum over the vasal vessels intact; preserve gubernacular vessels where possible.

LEVEL OF SPERMATIC VESSEL TRANSECTION

High (proximal) transection	Low (near testis) transection
Most surgeons transect at least 1.5–3 cm above the testis	Koff and Sethi proposed ligation close to the testis
Neheman: ligation as proximal as possible with a wide peritoneal flap in single-stage Fowler-Stephens, with very few failures	Rat studies: similar reduction in adult sperm numbers with high or low transection

- Vessel transection reduces germ cell counts, significantly in younger boys.
- Avoid vessel transection whenever possible — achievable in most abdominal orchidopexies.
- A retrovesical testis with a short vas may ultimately require orchidectomy.

SUCCESS OF ABDOMINAL ORCHIDOPEXY BY APPROACH

Approach	Success
Primary laparoscopic	>90% in most series; 97% in direct comparison
Primary open	89% in direct comparison; 95% for Jones and Bagley
Fowler-Stephens, overall	60–97%
One-stage Fowler-Stephens	63%
Two-stage Fowler-Stephens	68%
Microvascular autotransplantation	96% standard; 88% laparoscopically assisted
Shehata staged traction	84–85%; atrophy 10%

- Primary procedures are more consistently successful than Fowler-Stephens.
- Meta-analysis slightly favours two-stage over one-stage Fowler-Stephens (low-quality studies); recent data show equal atrophy but perhaps better position with two stages.
- Variation reflects age, position, testicular quality, follow-up and the definition of success ("intrascrotal" versus "dependent scrotal").
- Primary orchidopexy without vessel transection is preferable whenever possible.
- Some confirm viability by postoperative ultrasonography.
- Rare laparoscopic complications: bladder or vascular injury, hypercapnia, delayed small bowel obstruction.

Special applications and newer techniques

- Laparoscopy suits bilateral orchidopexy, abdominal wall defects, polyorchidism, splenogonadal fusion and TTE ± persistent müllerian ducts.

- **Simultaneous bilateral abdominal orchidopexy is common**, but **stage if both testes are very high or viability is doubtful**, choosing the contralateral approach at 6 months to **avoid bilateral atrophy**.
- **Microvascular autotransplantation** suits the **solitary abdominal testis** — **preserves the testis at the cost of operative time, microsurgical expertise and admission**; plastic surgeons can assist.
- **Monopolar coagulation of vessels as proximal as possible matches clip ligation histologically** in first-stage Fowler-Stephens.
- **Gubernaculum-sparing laparoscopic orchidopexy**: after first-stage Fowler-Stephens, **prominent vasculature arises from the gubernaculum in 60%**; **sparing it significantly lowers atrophy**.
- **Dessanti traction**: **PTFE (polytetrafluoroethylene) pericardial membrane wrapped round the cord** → **scrotal traction stitch** → at 9–12 months, **PTFE removed without reopening the canal and testis secured**; **up to 82% avoid reopening the canal**.

Shehata technique

- **Rationale**: **respiration and intra-abdominal pressure apply gradual traction to the gonadal vessels**, **lengthening them without ligation**.
- **Sequence**: **divide gubernaculum and incise lateral peritoneum** → **fix testis to the anterior abdominal wall medial to the contralateral anterior superior iliac spine** → **after 12 weeks, release and perform laparoscopic orchidopexy**.
- **Success 84–85%**, **higher in younger boys**; **atrophy 10%**.
- **Versus staged Fowler-Stephens**: **non-significantly better in a multicentre study**; **significantly higher success and lower atrophy in meta-analysis**.
- **No intestinal obstruction or hernia reported despite theoretical risk**.
- **Robot-assisted laparoscopic orchidopexy is feasible without clear advantage**.

Concept — every abdominal technique answers one question: **how to gain cord length without sacrificing the blood supply**. *Fowler-Stephens buys length by sacrificing the testicular artery; traction techniques buy it over time while keeping the artery* — *pushing vessel transection from first choice to last resort*.

KEY POINTS

- **Orchidopexy is recommended for testes undescended after 6 months of age and ideally before 18 months**. **Hormone therapy is not recommended**.

PROGNOSIS

- **Major gaps remain in understanding long-term outcome**, because surgery is expected to succeed and **prospective follow-up into adulthood is difficult**.
- **Adequate studies must account for confounders** — **severity (position, laterality, epididymal anomalies, hormones)**, **aetiology**, **age at surgery**, **hormone exposure**, **procedure** — and **use standardised methods with complete follow-up**.

Fertility

- **A history of cryptorchidism is strongly associated with subfertility**, particularly **bilateral but also unilateral**.
- **Cryptorchidism accounts for 10–17% of men evaluated for infertility**.
- **Individual prediction is difficult**: **diverse aetiology**, **age at treatment**, **approach and severity**.

- Interpretive limits: follow-up below one-third of eligible cases with self-selection, and reliance on a single semen analysis.

SEMEN OUTCOME AFTER CRYPTORCHIDISM — LARGE RETROSPECTIVE REVIEW

	Oligospermia	Azoospermia
Former bilateral	42%	31%
Former unilateral	31%	14%

- No difference by treatment before or after 9 years, and no effect of hCG, in that review.
- Azoospermia in unilateral disease reflects abnormal contralateral germ cell development; frequency is 0–10% in unselected cryptorchid series versus 11–28% in infertility clinics.
- Even after first-year orchidopexy, 20–25% remain at infertility risk on hormonal and biopsy criteria.
- Landmark population study (Schneuer et al., 2018): surgery before 18 months reduced the need for assisted reproductive technology (ART); each 6-month delay increased ART need by 5%.

SEMEN OUTCOMES ACROSS MAJOR SERIES

Series characteristic	Unilateral	Bilateral
Normal semen, surgery at mean 6.4 years	33%	18%
Normal sperm count, surgery at mean 7.0 years	60%	18%
Very low count (<5 million/mL), surgery at median 10.8 / 12.7 years	9%	54%
Normal concentration, surgery at 6–12 years	31%	—
Normospermia, surgery after puberty (14–29 years)	16%	—
Normal density (three analyses)	72% after orchidopexy; 77% after orchidectomy; 42% untreated	0%
Normal density, surgery at ≥7 years	84%	50%
After orchidectomy for unilateral disease (81% atrophic or Sertoli cell-only)	61% normospermic; 5% azoospermic	—
Normal count, surgery before 2 years	96%	75%

- Bilateral is always worse than unilateral, and the best results come from the earliest-operated series.
- Within series, outcome did not consistently track hormone therapy, adult hormones, position or age at surgery — but median surgical age was usually 6 years or more.
- Fertility is determined chiefly by germ cell quality in the descended testis in unilateral cases, and by unmeasured factors in bilateral cases.

Histological and hormonal predictors

- Mean germ cell counts correlate with semen parameters in some series but not others; individual predictive value is limited.
- Ad spermatogonia number is the better predictor.

AD SPERMATOGONIA AT BIOPSY AND ADULT SPERM COUNT

Ad spermatogonia status	Adult outcome
Present in both testes (no hormones)	Normal total count (>40 million/ejaculate) in 84%
Absent	Subnormal in all

Ad spermatogonia status	Adult outcome
Bilateral disease, present in one or both testes	Normal in 55%

- In a larger series, severe germ cell loss or abnormal Ad counts predicted lower mean counts and higher FSH, but group means stayed normal or non-significant — Ad analysis does not reliably predict the individual.
- Serum INHB and G/T are lower in bilateral than unilateral disease; in bilateral cases aged ≤ 24 months, G/T correlates moderately with INHB, AMH and the INHB/FSH and AMH/FSH ratios, and low ratios with high FSH predict impaired G/T — not seen in unilateral disease; hormones may aid counselling in bilateral cases.
- Ad spermatogonia count and germ cell absence are potentially useful prognostic measures, more so in bilateral cryptorchidism; larger prospective studies in infancy-operated boys are needed.

Limits of semen analysis

LANDMARK STUDY — GUZICK ET AL. (2001) SEMEN THRESHOLDS

Parameter	Subfertile	Indeterminate	Fertile
Concentration	$<13.5 \times 10^6/\text{mL}$	$13.5\text{--}48 \times 10^6/\text{mL}$	$>48 \times 10^6/\text{mL}$
Motility	$<32\%$	—	—
Normal morphology	$<9\%$	—	—

- Semen parameters overlap widely between men with and without proven paternity; thresholds are lower than the then-current World Health Organization criteria; about 3% of fertile men have density below $10 \times 10^6/\text{mL}$.
- Repeated analyses are needed in normal men, rarely obtained in cryptorchidism studies.
- In Tanner V youths under 25 years, samples were not highly reproducible but intraclass correlation showed substantial reliability, especially for total count — a single normal sample may suffice; repeat abnormal samples.

Paternity

- Paternity may be the more useful measure, limited by paternal discrepancy, female factors and variable interest in paternity.
- Paternal discrepancy: median 3.7% (interquartile range 2–9.6%).
- Pooled retrospective cohorts: paternity in about four-fifths of formerly unilateral and one-third of formerly bilateral men.

LANDMARK SERIES — LEE ET AL., PATERNITY AFTER ORCHIDOPEXY (1955–1975) VERSUS MATCHED CONTROLS

	Former bilateral	Former unilateral	Controls
Fathered children (married or cohabiting)	36%	53%	58%
Successful when attempting	65%	90%	93%
Semen	Density $\leq 13 \times 10^6/\text{mL}$ in all studied — yet some fathered children	Normal density and motility in 83%	—

- Unilateral did not differ from controls; attempt rates and lifestyle factors did not differ between groups.
- Paternity after orchidectomy for unilateral disease equalled controls.

- hCG treatment increased infertility risk (RR 4.7); testicular position and age at orchidopexy did not.
- Infertility is hard to predict without azoospermia or severe oligospermia.

UNILATERAL VERSUS BILATERAL CRYPTORCHIDISM — SEMEN AND PATERNITY

	Unilateral	Bilateral
Semen	Frequently abnormal	Usually abnormal
Paternity	Close to controls	Substantially reduced
Interpretation	Semen overstates real infertility	Semen reflects real infertility

Retractile testes

- Untreated childhood retractile testes: high paternity and normal adult volume in a small series.
- Infertility clinic series report poor semen after retractility, but retractility was poorly documented.
- Biopsy series found germ and Sertoli cell abnormalities in retractile testes qualitatively similar to cryptorchidism, some without controls.
- Retractable cannot be separated from acquired undescended testes in these studies.
- There is insufficient evidence of increased infertility risk in uncomplicated retractile testis.

Acquired cryptorchidism

- Data are retrospective, non-randomised and mix spontaneous pubertal descent with orchidopexy.
- Bilateral testes descending spontaneously after 10 years: volume <15 mL in most and count <20 million/mL in almost half.
- Dutch cohorts: no semen or hormone differences between spontaneous pubertal descent and pubertal orchidopexy (groups dominated by high scrotal and likely retractile testes).

SEMEN OUTCOME — CONGENITAL VERSUS ACQUIRED CRYPTORCHIDISM

	Congenital	Acquired
Oligospermia — unilateral	33%	39%
Oligospermia — bilateral	57%	71%
Azoospermia — unilateral	None	8%
Azoospermia — bilateral	None	7%
Significant difference	None — nor between observed and operated acquired cases	

- These data support a shared aetiology of congenital and acquired cryptorchidism.

Sperm retrieval

- Orchidopexy before 19 years gives significantly higher testicular sperm retrieval rates than after 19 years.
- Azoospermic men after orchidopexy have higher retrieval rates than those with idiopathic non-obstructive azoospermia.
- A congenital cause affecting both testes may not be abrogated by surgery alone, though early surgery is preferred and may help.

Concept — in unilateral disease the normal testis carries the fertility, and in bilateral disease nothing does.

This explains abnormal semen with near-normal paternity in unilateral men, unchanged paternity after orchidectomy of the affected side, and why every predictor — hormones, Ad spermatogonia, germ cell absence — works best in bilateral cases.

Testicular Cancer

- Cryptorchidism carries a two- to five-fold increased TGCT risk (pooled OR about 4).
- About 1% of cryptorchid cohorts develop TGCT; prior cryptorchidism is present in 5–10% of TGCT patients.
- Shared risk factors: inguinal hernia, prematurity, low birth weight.

INDEPENDENT RISK FACTORS FOR TGCT — SWEDISH POPULATION STUDY

Risk factor	OR	Age effect
Cryptorchidism	3.2	Stronger if diagnosed after puberty (~5.0) than before (~2.8)
Hypospadias	2.3	Stronger with younger presentation
Other genital malformation	1.9	—
Inguinal hernia	1.3	Stronger with younger presentation

- Meta-analysis confirms congenital cryptorchidism approximately triples adult testicular cancer risk.
- Contralateral descended testis: OR about 1.5–1.7, the lower estimate non-significant.
- Landmark meta-analysis (Walsh et al., 2007): orchidopexy after 10–11 years carries RR 5.8 versus earlier correction.

AGE AT ORCHIDOPEXY AND TGCT RISK (OSTERBALLE ET AL., 2017)

Age at orchidopexy	OR
Under 9 years	1.1–2.35
10–14 years	2.9
13–15 years	5.1
16 years and over	6.2
Untreated adults	14.4

- In that long-term cohort (mean surgery 10.8 years, follow-up ~29 years), TGCT arose in about 1% at mean age 33, equally unilateral and bilateral, a minority in the contralateral testis; most followed orchidopexy at ≥10 years; prepubertal intratubular neoplasia was rare and mostly syndromic, including DSD.
- Cancer risk rises with every 6 months' delay beyond 18 months of age.

Germ cell neoplasia in situ

- Carcinoma in situ, intratubular germ cell neoplasia unclassified (IGCNU) and germ cell neoplasia in situ (GCNIS) denote the same precursor.
- Prevalence: 2–3% in previously cryptorchid men; 0–0.4% in children with nonsyndromic cryptorchidism.
- Contralateral GCNIS in 4–8% of men with TGCT; predictors are testicular atrophy, cryptorchidism, and age under 40.

TUMOUR HISTOLOGY BY TREATMENT STATUS

Seminoma	Nonseminomatous germ cell tumour (NSGCT)
Associated with persistently cryptorchid testes (74%)	Predominates in testes corrected by orchidopexy (63%)

- Both arise from GCNIS, which forms from gonocytes that fail to differentiate and undergo adaptive genetic and epigenetic change.

- **Failure of PGCs to differentiate may produce infantile teratoma/teratocarcinoma and yolk sac tumour** — rare, reported in prepubertal cryptorchid boys.
- **Multinucleated spermatogonia occur in 8% of orchidopexy biopsies**, in younger boys with normal or high germ cell counts; **significance unknown**.

Biopsy and orchidectomy

- **Risk is higher with chromosomal defects or DSD.**
- **Husmann recommends biopsy in these and in boys older than 12 years at orchidopexy**; the age cut-off is not clearly defined.
- **Orchidectomy is preferred for cryptorchid testes in postpubertal men up to 50 years.**

DOES BIOPSY AT ORCHIDOPEXY RAISE TGCT RISK?

Increased risk	No increased risk
Retrospective cohort RR 6.7 — biopsy indications unknown, so confounding by indication likely	Large Scandinavian cohort with universal biopsy: no excess beyond that expected

Concept — cancer risk is a function of how long the gonocyte defect sits in an abnormal environment, layered on an intrinsic defect surgery cannot remove. *The age dose-response and seminoma excess in uncorrected testes are the modifiable component; the contralateral excess and residual risk after early surgery are the intrinsic one.*

Testicular microlithiasis

- **Multiple specular calcifications within the testicular parenchyma; commoner with GCNIS or TGCT, but present in the normal population and at similar frequency after cryptorchidism.**

PREVALENCE OF TESTICULAR MICROLITHIASIS

Population	Prevalence
Normal population, general estimate	5–10%
Population-based, primarily White	4.2%, rising with age; half limited (<5 lesions per testis), not clinically significant
Multicentre review, >37,000 males	2.9%
Younger boys with cryptorchidism	3.5%
Older males followed for acquired cryptorchidism	6.4%
Previously cryptorchid — affected vs contralateral	14% vs 19%, no difference; none between congenital and acquired

- **Figures differ with age structure, ethnicity and the limited-versus-classic definition.**
- **Tumours in 4.64% with versus 0.33% without microlithiasis, but cryptorchidism was not addressed.**
- **Among boys followed for microlithiasis, only one developed malignancy during puberty.**
- **Microlithiasis is not clearly an independent TGCT risk factor; no data show added risk with cryptorchidism; follow-up is undefined.**
- **Rete testis ectasia and intratesticular varicocele are also seen after cryptorchidism; significance undefined. Routine ultrasonography is not recommended.**

Surveillance and self-examination

- **The U.S. Preventive Services Task Force grades routine testicular self-examination (TSE) "D", partly because TGCT treatment is highly effective — but not explicitly for previously cryptorchid men.**

- **Cost-benefit analyses favour recommending TSE.**
- **Counsel patients and families on infertility and TGCT risk and the potential benefit of TSE.**

Concept — surveillance rests on counselling and self-examination, not imaging. *Microlithiasis and other sonographic findings have no defined significance, so the evidence-aligned approach is informed self-examination in a population whose absolute risk is modest but real.*

KEY POINTS

- **Sperm counts are reduced in at least 25% of formerly unilateral and the majority of formerly bilateral cryptorchid men; paternity in the unilateral group is similar to controls.**
- **Ad spermatogonia counts may predict fertility potential in some males with cryptorchidism.**
- **TGCT risk is two to five times higher with cryptorchidism and rises with age at surgery.**

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